

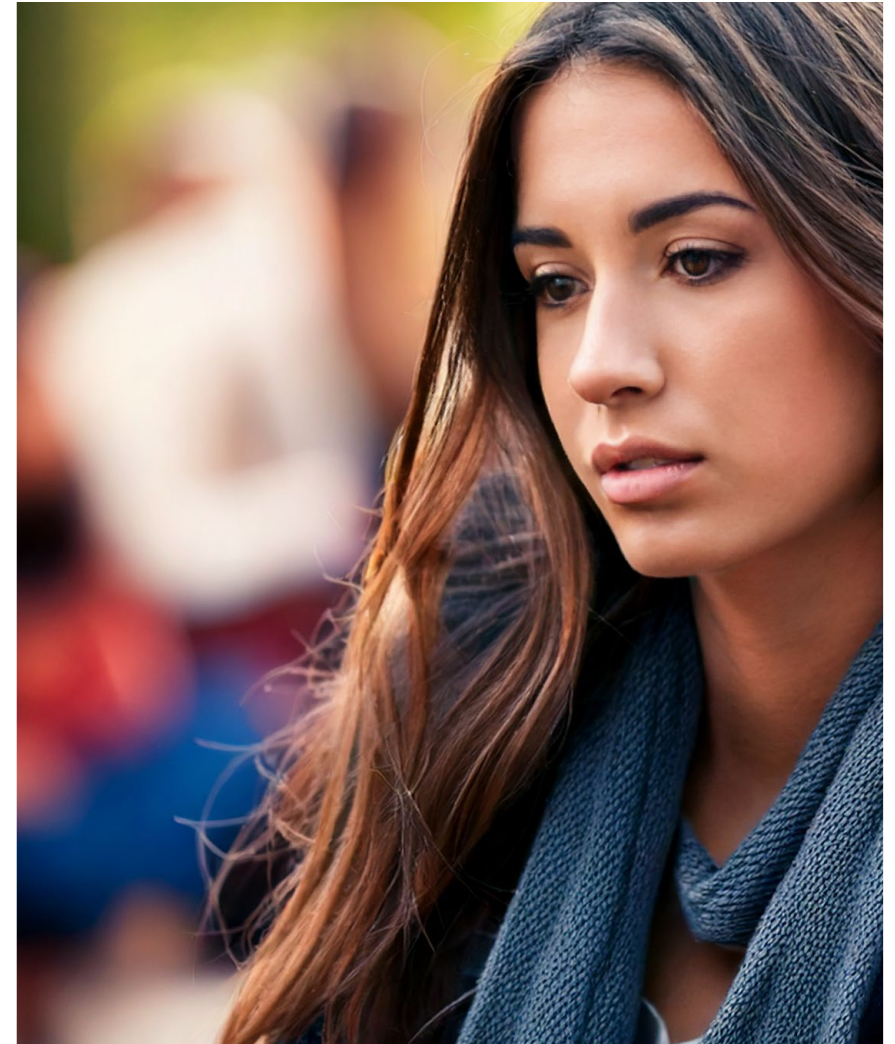
Loneliness Moderates the Association Between Sensory Processing Sensitivity & Health

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SPS and Social Challenges

- Linked to social and communication deficits, social phobia, lower depth of relationships, and less positive relations with others
- Distinct from introversion & shyness
- Low sociability may serve as a coping mechanism to avoid being overwhelmed by sensory experience



BACKGROUND



SPS and Loneliness

- Based on a 16-item *Sensory Processing Sensitivity Questionnaire*, SPS associated with greater emotional loneliness

SPS and Preference for Solitude

- Qualitative studies: need for solitude
- SPS positively correlated ($r = .40$) with self-determined solitude



BACKGROUND



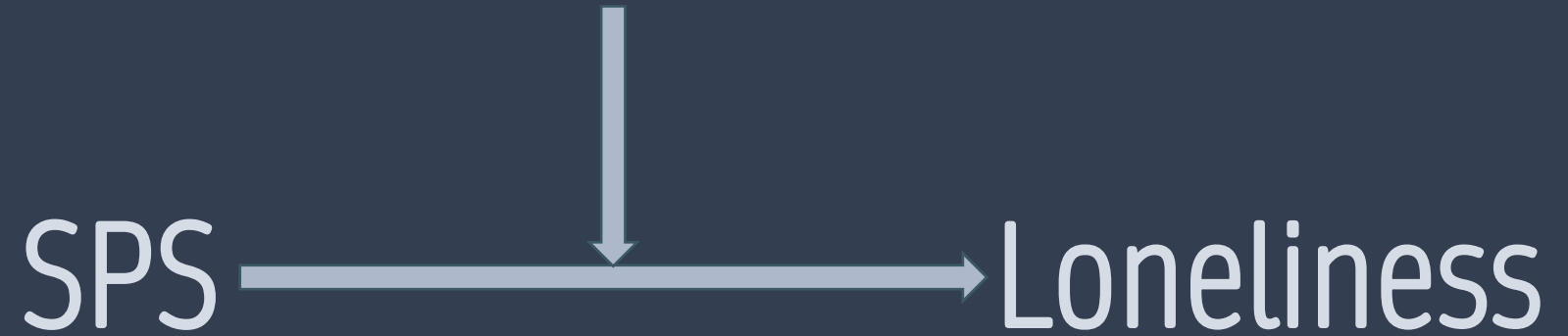
SPS → Loneliness

H1. SPS associated with greater loneliness

BACKGROUND



Preference for Solitude



H2. SPS-loneliness moderated (**attenuated**) by preference for solitude



- A growing body of work, including studies from my lab and others, suggests that SPS is linked to physical health issues.

- Comprehensive review/meta-analysis demonstrated medium to large effects on variety of health outcomes



BACKGROUND



H3. SPS associated with poorer physical health

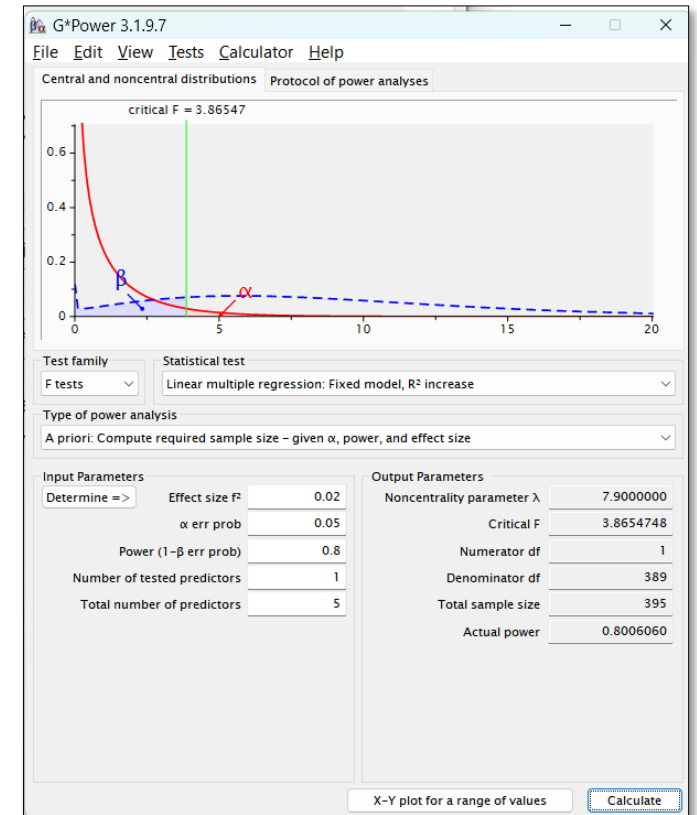
BACKGROUND

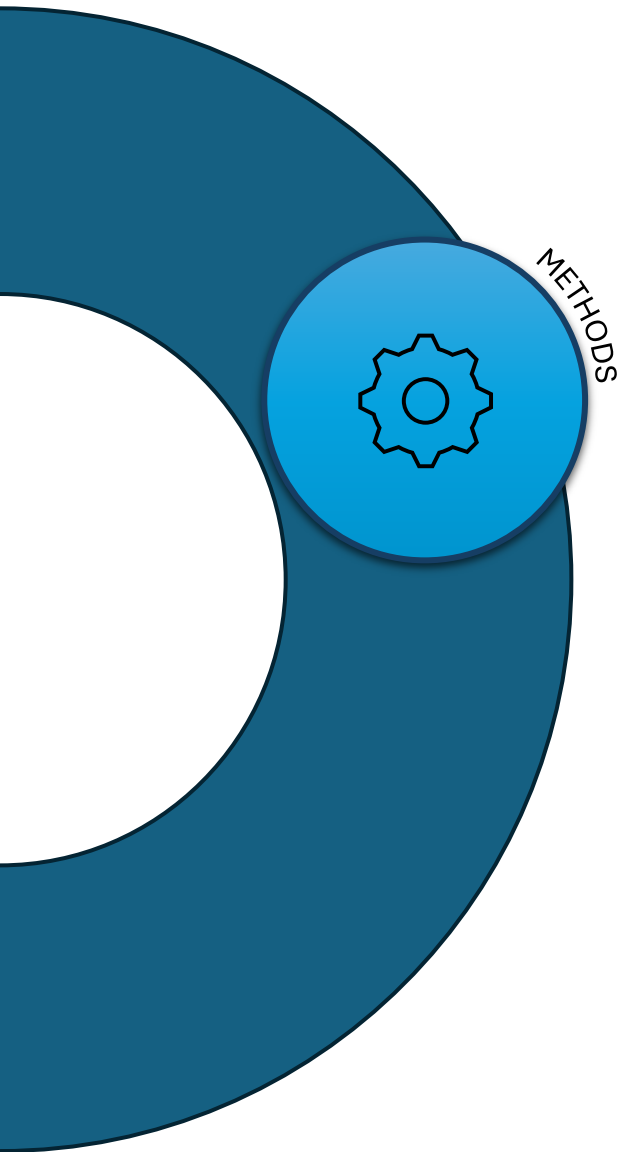


H4. SPS-health moderated (**amplified**) by loneliness

Power Analysis

- Estimated sample size = **395**
- Effect size = .02 (small)
- Power = .8



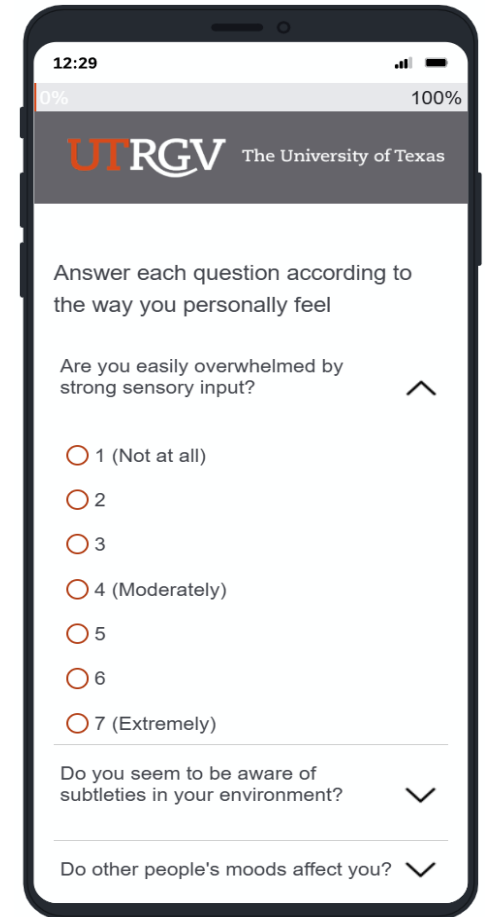


Sample

- 449 undergraduates
 - Mage = 20.5; 64.1% female; 94.2% Hispanic

Online survey

- Highly Sensitive Person Scale (27-items)
- UCLA Loneliness Scale – 10 (10 items)
- Preference for Solitude Scale (12 items)
- Cohen-Hoberman Inventory of Physical Symptoms (33 items)



Data Analysis

METHODS

- Custom R Shinyapp for moderation
 - Standardized residual threshold = 2.0 (~30 cases removed)
 - Hayes' PROCESS (5,000 bootstrapped samples, controlling for *age*, *sex*, and *income*).

The screenshot shows the 'Moderation Analysis with PROCESS for R' Shiny application. The interface is divided into two main panels: a left sidebar for settings and a right panel for results.

Left Panel (Settings):

- Upload Data:** 'Choose CSV or SAV File' with a 'Browse...' button and a status 'SSH 2024 12-4-2024 - 449s'. An 'Upload complete' button is visible.
- Select Variables:**
 - Outcome Variable:** 'chip_sum' (dropdown)
 - Predictor Variable:** 'HSP_mean' (dropdown)
 - Moderator Variable:** 'ULS10_sum' (dropdown)
 - Covariates (optional):** 'Age_corrected', 'Sex', 'Individual_Income' (checkboxes)
 - Standardized Residual Threshold:** '2' (input field)
 - PROCESS Settings:**
 - ☒ Use bootstrapping
 - Number of bootstrap samples:** '5000' (input field)
 - Centering:** ☒ No centering, ☐ Mean center, ☐ Standardize
 - Conditioning Values:** ☒ Mean and ± 1 SD, ☐ 16th, 50th, 84th percentiles
 - Heteroscedasticity-consistent inference:** (checkbox)

Right Panel (Results):

- Assumption Checks:** (tab)
- Moderation Analysis:** (tab)
- Analysis Results:**
 - ANALYSIS SUMMARY:**
 - Dataset name: SSH 2024 12-4-2024 - 449s - nonresponders deleted - main measures completed
 - Dataset size: 449 cases
 - 30 standardized residual outliers ($|SR| > 2.0$) removed
 - Predictor variable: HSP_mean
 - Outcome variable: chip_sum
 - Moderator variable: ULS10_sum
 - Covariate(s): Age_corrected, Sex, Individual_Income
 - ANALYSIS SETTINGS:**
 - Configuration used for this analysis:
 - Centering: No centering
 - Bootstrap samples: 5000
 - Conditioning values: Mean and ± 1 SD
 - Heteroscedasticity-consistent SE: MCP (Huber-White)
 - MODEL FIT STATISTICS:**
 - This section shows how well the overall model fits the data:
 - R-squared indicates the proportion of variance explained
 - F-test shows if the model is significantly better than no predictors
 - Model Summary:**

	R	R-sq	NSE	F	df1	df2	p
	0.5993	0.3592	305.9636	77.5437	3.0000	415.0000	0.0000

Exploratory Factor Analysis

- Custom R Shinyapp for automated iterative EFA
 - Multicollinearity check: HSP items 1 and 14 removed (determinant threshold)
 - Spearman correlation matrix used (multivariate non-normality)
 - Good sampling adequacy (Bartlett sphericity, KMO)

Integrated gbEFA Application

Introduction Multicollinearity Check Factor Estimation Diagnostics **Exploratory Factor Analysis**

Exploratory Factor Analysis

Upload CSV File

Browse... SSH 2024 12-4-2024 - 449s - nonresponders deleted - main measures - **Upload complete**

☒ Dataset has a header row

Number of Factors

2

Maximum recommended factors: 5 Based on your data having 25 variables. Note: Maximum is limited to ensure stable results.

Factor Extraction Method

Principal Axis Factoring

Rotation Method

promax

Correlation Matrix Type

spearman

Note: Polychoric correlations are recommended for ordinal data (e.g., Likert scales)

Loading Threshold

0.3

Communality Threshold

0.2

Scale Minimum Value (observed): 1

Overall Summary

Dataset Information:

- Dataset name: SSH 2024 12-4-2024 - 449s - nonresponders deleted - main measures completed HSP only for EFA - V1 and V14 removed V2.csv
- Number of rows of data: 449
- Report generated: 2024-12-31 23:35:03

Analysis Settings:

- Number of Factors: 2
- Rotation Method: promax
- Factor Method: pa
- Correlation Method: spearman
- Loading Threshold: 0.3
- Communality Threshold: 0.2

Analysis Results:

- Total Iterations: 3
- Initial Variables: 25
- Final Variables: 18

Reliability Statistics

Overall Scale Reliability:

- Cronbach's alpha: 0.917
- McDonald's omega: 0.931

Factor-Specific Reliability:

	Variables	% of Variance Explained	Cronbach's alpha	McDonald's omega
PA1	HSP_4, HSP_5, HSP_7, HSP_9, HSP_11, HSP_13, HSP_19, HSP_20, HSP_21, HSP_23, HSP_25, HSP_26	30.5	0.912	0.928
PA2	HSP_2, HSP_10, HSP_12, HSP_15, HSP_17, HSP_22	13.1	0.790	0.850

Bivariate Correlations

	SPS	Loneliness	Preference for Solitude	Physical Symptoms	Age	Sex
Loneliness	0.30					
Preference for Sol.	0.31	0.25				
Physical Symptoms	0.43	0.26	0.19			
Age	0.09	0.09	0.14	0.04		
Sex	0.20	-0.01	0.14	0.20	0.02	
Income	0.05	0.08	0.07	0.05	0.50	-0.04



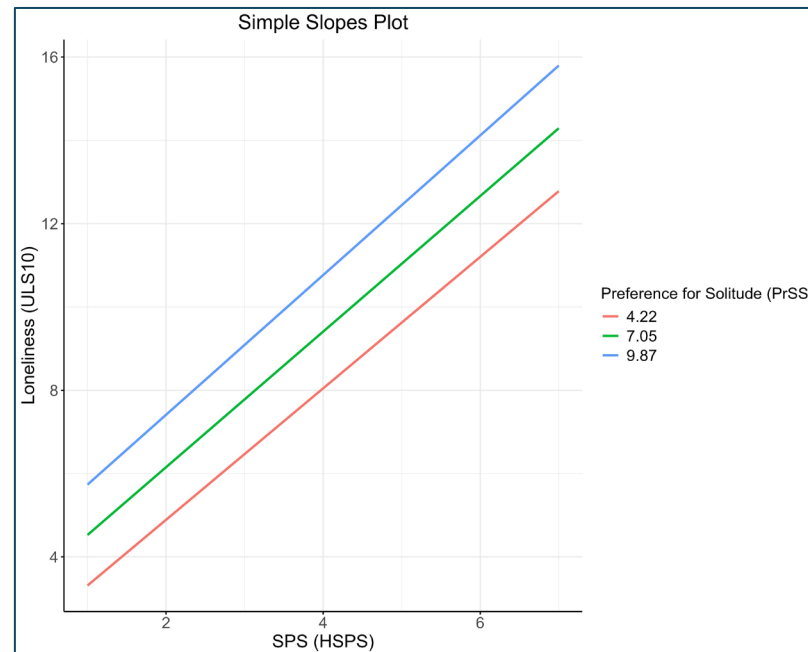
RESULTS

SPS - Loneliness

- H1. SPS was positively correlated with loneliness ($r(447) = .30$).
- H2. Preference for solitude did not moderate the association ($F(1, 426) = 0.05, p = .823$).

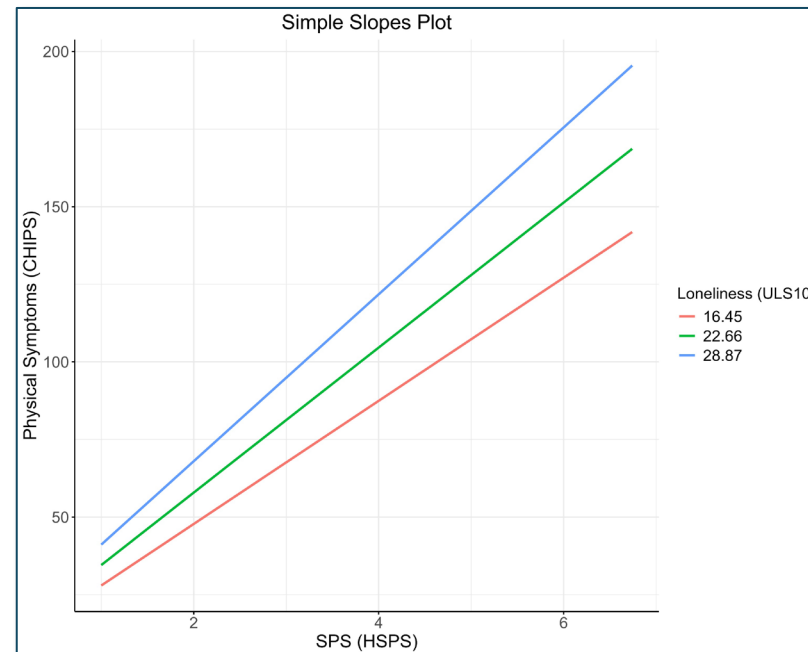
SPS → Loneliness

Preference
for Solitude
↓
SPS → Loneliness



SPS - Health

- H3. SPS was positively correlated with physical symptoms ($r(447) = .43$)
- H4. Association stronger in those with greater loneliness ($F(1, 415) = 19.97, p < .001$).

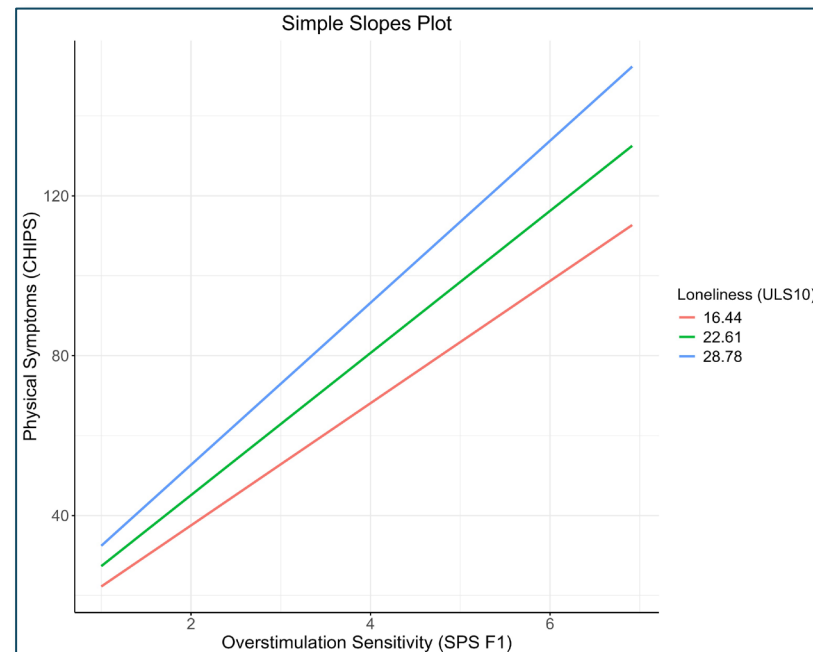


SPS Factors - Health

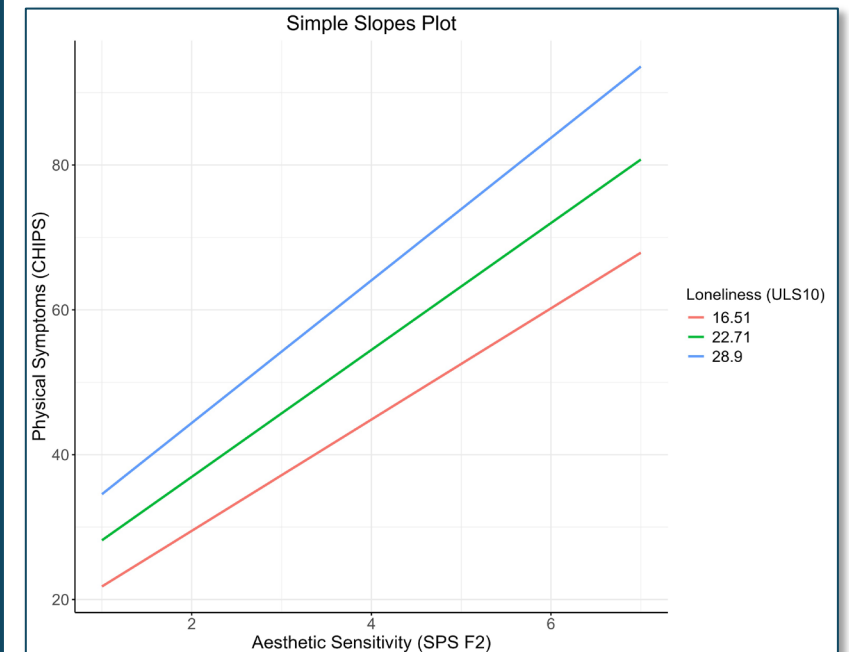
- EFA revealed indicated a two-factor solution for SPS:
“*overstimulation sensitivity*” (OSS) and “*aesthetic sensitivity*” (AES).

	OSS (F1)	AES (F2)	Physical Symptoms
OSS (F1)	1.000		
AES (F2)	.592*	1.000	
Physical Symptoms	.471*	.205*	1.000
Loneliness	.327*	.148*	.259*

OSS-Health association stronger in those with greater loneliness ($F(1, 414) = 16.45, p < .001$).



AES-Health association not moderated by loneliness ($F(1, 421) = 1.48, p = .224$).



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CONCLUSION

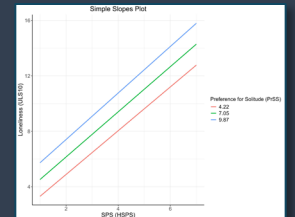


Preference for Solitude

- Black & Kern: *“...interviewees reported that regular experiences of solitude were an important enabler of their wellbeing.”* (2020, p. 6).
- Although we hypothesized that a preference for solitude would attenuate the SPS – loneliness association, this was not the case.
- Recent research suggests a **preference for solitude paradox**: it does not offer a protective effect for poor mental health in those who are socially isolated.

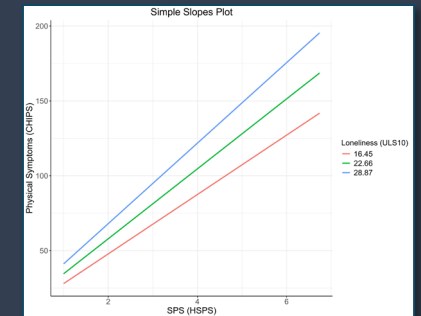
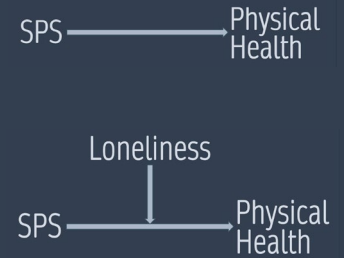


CONCLUSION



SPS - Health

- Replicating prior research, SPS associated with poorer health. Importantly, loneliness appears to amplify this association.
- Viewing this another way, the association between loneliness and health may be greater in those with higher SPS scores. SPS may be an important individual difference consideration for future studies on the impact of loneliness on health.



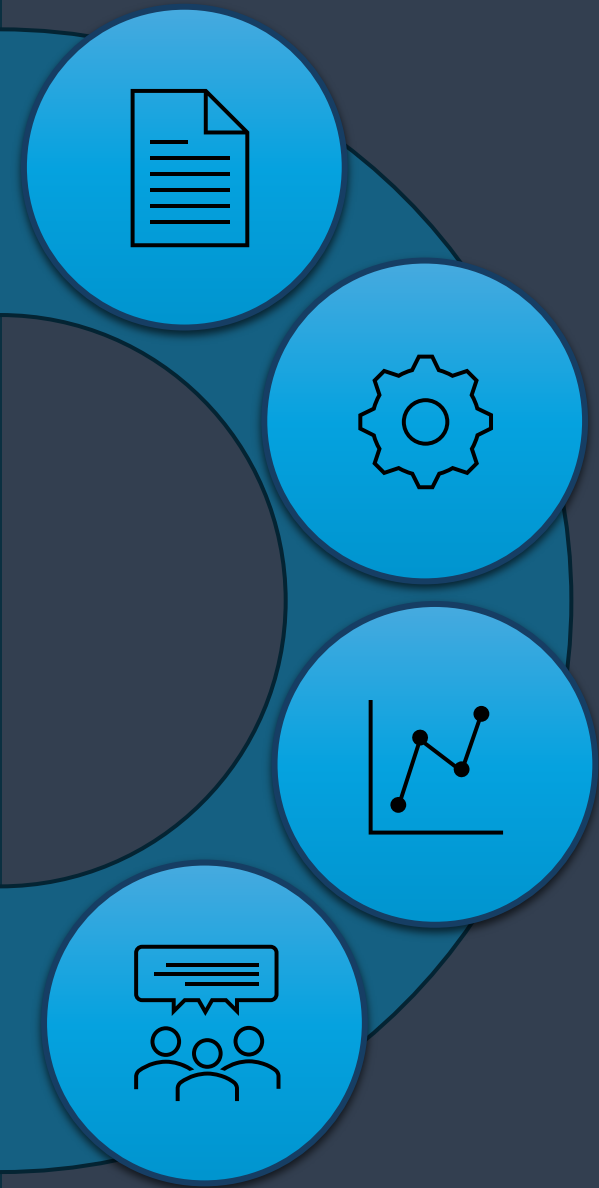
CONCLUSION

Limitations

- Sample limited to young adults, most of whom were Hispanic.
- Physical health measures were based on subjective self-report.
- The potential impact of positive or negative childhood experiences were not examined.
- Although loneliness and preference for solitude are subjective by their very nature, objective measures of time spent alone or with others, along with the activities engaged in during these times and the quality of relationships, might be important factors.



CONCLUSION



Questions **and** Answers